

ITEM 13

Practice 1

User: Diya Jain

In this exercise, you need to compute $107 + 9$ using 8 - bit 1's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary

Please convert 107 to signed decimal to 8-bit binary representation with 1's complement: 01101011

Please convert 9 to signed decimal to 8-bit binary representation with 1's complement: 00001001

Step 2a: Perform binary addition

```
  0 1 1 0 1 0 1 1
+ 0 0 0 0 1 0 0 1
-----
```

Please input the final carry bit (0 or 1)?: 0

Please input the above binary addition result: 01110100

```
  0 1 1 0 1 0 1 1
+ 0 0 0 0 1 0 0 1
-----
```

```
  0 1 1 1 0 1 0 0
```

Step 2b: End-Round Carry?

End-round carry operation is required for this binary addition? (y/n): n

Step 3: Verify the result.

Please convert the result binary sum to decimal value: 116

Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.

Well done!

Practice 2

User: Diya Jain

In this exercise, you need to compute $14 + 22$ using 8 - bit 1's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary

Please convert 14 to signed decimal to 8-bit binary representation with 1's complement: 00001110

Please convert 22 to signed decimal to 8-bit binary representation with 1's complement: 00010110

Step 2a: Perform binary addition

```
  0 0 0 0 1 1 1 0
+ 0 0 0 1 0 1 1 0
-----
```

Please input the final carry bit (0 or 1)?: 0

Please input the above binary addition result: 00100100

```
  0 0 0 0 1 1 1 0
+ 0 0 0 1 0 1 1 0
-----
```

```
  0 0 1 0 0 1 0 0
```

Step 2b: End-Round Carry?

End-round carry operation is required for this binary addition? (y/n): n

Step 3: Verify the result.

Please convert the result binary sum to decimal value: 36

Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.

Well done!

ITEM 14

Practice 1

Please select the practice item: 14

User: Diya Jain

In this exercise, you need to compute $2 + 116$ using 8 - bit 2's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary

Please convert 2 to signed decimal to 8-bit binary representation with 2's complement: 00000010

Please convert 116 to signed decimal to 8-bit binary representation with 2's complement: 01110100

Step 2: Perform binary addition

0 0 0 0 0 0 1 0

+ 0 1 1 1 0 1 0 0

Please input the final carry bit (0 or 1)?: 0

Please input the above binary addition result: 01110110

0 0 0 0 0 0 1 0

+ 0 1 1 1 0 1 0 0

0 1 1 1 0 1 1 0

Step 3: Verify the result.

Please convert the result binary sum to decimal value: 118

Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.

Well done!

Practice 2

Please select the practice item: 14

User: Diya Jain

In this exercise, you need to compute $77 + 25$ using 8 - bit 2's complement operation. Verify if the result is correct.

Step 1: Convert decimal data to binary

Please convert 77 to signed decimal to 8-bit binary representation with 2's complement: 01001101

Please convert 25 to signed decimal to 8-bit binary representation with 2's complement: 00011001

Step 2: Perform binary addition

0 1 0 0 1 1 0 1

+ 0 0 0 1 1 0 0 1

Please input the final carry bit (0 or 1)?: 0

Please input the above binary addition result: 01100110

0 1 0 0 1 1 0 1

+ 0 0 0 1 1 0 0 1

0 1 1 0 0 1 1 0

Step 3: Verify the result.

Please convert the result binary sum to decimal value: 102

Overflow occurs or not (y/n)?: n

The final result is correct and no overflow occurs.

Well done!